2019 CATHOLIC HIGH SCHOOL CALCULUS B INDIVIDUAL

1. Find $\lim_{x \to \frac{\pi}{4}} \left(\frac{\sin x - \cos x}{\tan x - 1} \right)$

2. Find $\lim_{h \to 0} \left(\frac{7^{x+h} - 7^x}{h} \right)$

3. Find the slope of the tangent to $xy + 2x - 5y = 2$ at $(3,2)$

4. If $f(t) = e^{2t} \sin(3t)$, find $f'(0)$

5. If $u = \ln\sqrt{v^2 + 2v - 1}$, find and simplify $\frac{du}{dv}$

6. Find $\frac{dy}{dx}$ if $y = \frac{1 + \sin x}{1 - \sin x}$

7. Find $\frac{d^3y}{dx^3}$ if $y = \ln(5x)$

8. Find all critical point(s) for $f(x) = (x-2)(x-3)^4$

9. If $f(x)=x-2\sin x$ on $[0,2\pi]$, give the x coordinate(s) of the relative maximum point(s)

10. At what point on $y=(\ln(x+4))^2$ is the tangent line horizontal?

11. As a spherical balloon is being inflated, its radius r (in cm) is given by $r=3\sqrt[3]{t}$ for $0\le t\le 10$. Find the rate of change after 8 seconds for the volume of the balloon (include units)

12. Find the area of the region bounded by $y=\frac{1}{\sqrt{1-x^2}}$ and $y=-\frac{1}{\sqrt{1-x^2}}$ on $\left[-\frac{1}{2},\frac{1}{2}\right]$

13. Find y if $\frac{dy}{dx}=24x(3x^2-1)^3$ and $y(0)=-3$

14. $\int_{\sqrt{e}}^{e}\left(\frac{\ln x}{x}\right)dx$

15. Find $\int \sin^2(x)\cos^3(x)\,dx$

Catholic High Solutions

1. $\sqrt{2}/2$
2. $7^x \cdot \ln 7$
3. 2
4. 3
5. $\frac{v+1}{v^2+2v-1}$
6. $\frac{2\cos x}{(1-\sin x)^2}$
7. $2x^{-3}$
8. $(3,0)$, $(\frac{11}{5}, \frac{4^4}{5^5})$
9. $x = 5\pi/3$
10. $(-3,0)$
11. 36π cm^3/sec
12. $2\pi/3$
13. $y = (3x^2-1)^4 - 4$
14. $3/8$
15. $\frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + c$